UJJAWAL PRATAP SINGH

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EDUCATION

University of Alberta, Faculty of Science, Edmonton, AB May 2027
B.Sc Computing Specialization Program
Relevant Coursework: Intro to Software Engineering, File and Database Management, Intro to Machine Learning, Reinforcement Learning, Computer Architecture, Data Structures & Algorithms, Practical Programming Methodology, Formal Systems and Logic, Web Applications and Architecture

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3, C, Bash, RISC-V
Web & Backend: React 18, Node.js, FastAPI, TailwindCSS, WebSocket/SSE, Firebase Firestore, REST APIs
Machine Learning: Generative AI, LLM Orchestration (Gemma, DeepSeek), RAG, NLP, PyTorch, Reinforcement Learning
DevOps & Tools: Docker, CI/CD (GitHub Actions), Git/GitHub, Linux/Unix, VS Code, Role-Based Access Control (RBAC)
Data & Analytics: MongoDB (NoSQL), PostgreSQL, Query Optimization, Data Pipelines, JSON Processing

PROJECTS

Leo the Lion | AI-Powered Pediatric Health Platform (Hackathon Top 5)

- **Full-Stack Architecture:** Architected a real-time health companion using **React 18 (Frontend)** and **FastAPI (Backend)**, featuring a dual-interface system for children and parents**.**
- **AI & RAG Integration:** Engineered a **Retrieval-Augmented Generation (RAG)** engine using **Python** and Sentence Transformers to ground responses in medical protocols, utilizing **Google Gemma 3** for empathetic dialogue.
- **DevOps & CI/CD:** Containerized the application using Docker for consistent deployment and implemented automated pipelines, aligning with CI/CD best practices**.**
- **Real-Time Optimization:** Engineered a low-latency bi-directional streaming architecture using **Server-Sent Events (SSE)**, reducing perceived latency by 40% for instant voice interaction.

Event Lottery System (Android, Java, Firebase)

- **Scalable Cloud Architecture:** Engineered a real-time event management system using **Firebase Firestore (GCP** ecosystem), handling concurrent user data and live profile synchronization**.**
- **Algorithmic Fairness:** Implemented a bias-free randomization algorithm to process waitlists and select entrants, ensuring mathematical fairness for high-demand events.
- **Security & Access Control:** Architected a **Role-Based Access Control (RBAC)** system to enforce secure permissions between Entrants, Organizers, and Admins.

MongoDB Data Analytics Interface (Python, NoSQL)

- **Data Pipeline Optimization:** Engineered a high-performance data ingestion pipeline for large datasets using batched JSON processing, significantly reducing processing time **for large-scale data cleaning and ingestion tasks**.
- **Advanced Analytics & Trends:** Constructed complex aggregation pipelines within MongoDB to calculate "helpfulness ratios" and identify top-tier reviewer trends, simulating **Voice of Client (VoC)** sentiment analysis logic.
- **Data Integrity:** Developed logic to automatically flag suspicious review patterns and analyze trends across diverse product categories, ensuring a quality data foundation.

LEADERSHIP & AWARDS

Indian Students Association (INDSA), VP Media Jan 2025 - Sep 2025

- **GenAI Integration:** Integrated **Generative AI workflows** (LLMs, Image Gen) into the content strategy to accelerate drafting and design processes, reducing production time while maintaining brand consistency.
- **Multimedia Strategy:** Led the end-to-end digital branding strategy for one of the largest campus organizations, utilizing **AI-enhanced editing software** to produce high-fidelity content across multiple platforms.
- **Data-Driven Optimization:** Analyzed audience engagement metrics using **AI-powered analytics tools** to identify trends and optimize scheduling, directly resulting in a sold-out major cultural event.

Awards & Recognition

- 1st Place - Al Tasamuh National Inter-School Robotics
- Awarded $5000 for the University of Alberta Regional Excellence Scholarship
- Awarded $5000 for the University of Alberta International Admission Scholarship